FLARE: A Contract-Verified Benchmark for Risk-Aware UAV Planning Under Coupled Dynamic Hazards

Konstantinos Zervakis* Elias Panagiotopoulos* *Athens, Greece

***Abstract*—We present FLARE, an open-source benchmark for UAV path planning in disaster-response scenarios where planner decisions have *human consequences*. Three OpenStreetMap-derived urban scenarios—pharmaceutical delivery during wildfire, multi-casualty urban rescue, and fire-perimeter surveillance—feature wind-driven cellular-automaton fire, structural collapse with permanent debris, traffic closures, and multi-POI missions with time-decaying impact scores. A *risk cost map* translates hazard proximity into per-planner costs. Five planners are evaluated across 450 episodes (30 seeds). Beyond navigation success, we introduce *mission scores* that quantify medication efficacy, casualty survival, and surveillance freshness. These scores reveal ranking inversions: planners that reach the goal may score lower than those that prioritize task completion. A contract architecture (36 invariants, 15 families) guarantees deterministic replay and planner fairness. All experiments are bit-identical under the same seed triple. Code and data: `MIT license`.**

***Index Terms*—UAV path planning, disaster response, benchmarking, dynamic environments, fire spread, mission-impact scoring, reproducibility**

I. Introduction

When a UAV delivers insulin to a fire-isolated village, every minute of delay degrades the medication. When it triages fire-stranded casualties, the order and speed of visits determine who survives. Standard path-planning benchmarks reduce these missions to a binary: did the agent reach the goal? They cannot distinguish a planner that delivers insulin in 200 steps from one that arrives in 800—yet the patient outcomes are vastly different.

UAVs are increasingly deployed in disaster response [1], [2], [3], [4], where wildfires advance under wind and buildings collapse. Path planning algorithms must handle *dynamic* obstacles in rapidly evolving environments [5], [6]. Despite extensive research [7], [8], reproducible comparison under controlled dynamic conditions remains difficult: ad-hoc scenarios, fixed obstacles, and proprietary simulators [9] prevent isolation of environmental effects. Furthermore, existing benchmarks report only navigation success, ignoring the *mission impact* of planning decisions [10].

We address these gaps with **FLARE**, a benchmark framework with four design principles:

1) **Deterministic reproducibility.** Given the same $(scenario, planner, seed)$ triple, the framework produces bit-identical trajectories, event logs, and metrics. A single root RNG, spawned per episode, drives all stochastic components.
2) **Planner fairness.** All planners receive identical observations at each step. Dynamic obstacles evolve *after* the agent moves, and interdictions are placed on the A*-reference corridor independent of which planner is under evaluation.
3) **Hazard-coupled planning.** Wind-driven fire [11], continuous risk cost maps, and structural collapse create a rich hazard landscape that differentiates planners by their risk tolerance, not just their replanning frequency.
4) **Rigorous evaluation.** We adopt the statistical protocol of Agarwal et al. [10]: interquartile mean (IQM) with bootstrap confidence intervals, Friedman tests [12], Cliff's delta effect sizes, and performance profiles [13].

The framework implements a Gymnasium-compatible [14] environment with three urban scenarios built from OpenStreetMap data [15], each inspired by real Greek disaster events (2021 Evia wildfires, 2023 Rhodes coastal fires, 2023 Evros megafire). Five planners from three algorithmic families are evaluated: A* [16], Incremental A* (inspired by D* Lite [17]), APF [18], and two A*-based replanning variants.

Contributions.

- Multi-POI disaster-response missions with time-decaying impact scores (medication efficacy, casualty survival, surveillance freshness) that reveal ranking inversions invisible to navigation-only metrics.
- An open-source, contract-verified framework (36 invariants, 15 families) with wind-driven fire and per-planner risk costs.
- Three OSM-based urban scenarios with physically motivated dynamic obstacles and operationally grounded missions.
- A 450-episode study with rigorous statistical analysis, plus ablation studies isolating individual dynamics layers.

II. Related Work

A. UAV Path Planning

Classical UAV planners include A* [16], D* Lite [17], sampling-based methods (RRT, PRM), and APF [18]. Recent surveys [5], [6] emphasize benchmarks under dynamic, uncertain conditions. Risk-aware planning [19], [20] uses continuous cost maps from hazard proximity; FLARE implements per-planner risk coefficients for risk-tolerant vs. risk-averse comparison.

B. Benchmarking and Reproducibility

Henderson et al. [9] showed RL results are sensitive to seeds and evaluation methodology. Agarwal et al. [10]

proposed IQM with bootstrap CIs; Demšar [12] formalized Friedman/Wilcoxon tests; Dolan and Moré [13] introduced performance profiles. FLARE adopts all four. Gymnasium [14] provides the standard environment API; Bonsignorio et al. [21] argue for reproducible robotics benchmarks—a principle central to FLARE's contracts.

C. Dynamic Environments

Alexandridis et al. [11] proposed CA wildfire models; recent work [22], [23] confirms wind as the dominant spread parameter. FLARE extends this with directional modulation and smoke diffusion. Boeing [15] introduced OSMnx for rasterized urban terrain.

III. System Design

A. Overview

FLARE is a Python framework built around a single Gymnasium environment (`UrbanEnvV2`) with deterministic dynamics. A benchmark runner orchestrates episodes: it resets the environment with a seed, initializes the planner, and iterates the step loop until termination. The runner owns the authoritative step counter; all components (dynamics, logger, renderer) receive it as a parameter.

The architecture follows a layered design: *(i)* scenario configs define map, dynamics knobs, and mission parameters; *(ii)* the environment manages grid state, collision semantics, and dynamics stepping; *(iii)* planners implement a unified `PlannerBase` interface (`plan()`, `update()`, `should_replan()`); *(iv)* the runner enforces the step protocol and collects metrics.

Algorithm 1 Episode Runner

Require: scenario s, planner π, seed k
1: rng $\leftarrow$ RNG(k); $(o, \text{mask}) \leftarrow$ env.reset(s, rng)
2: π.init(o); $t \leftarrow 0$
3: **while** $t < T_{\max}$ **and not** done **do**
4: $a \leftarrow \pi$.act(o, mask) {mask frozen}
5: $(o, r, \text{done}, \text{info}) \leftarrow$ env.step(a)
6: {Dynamics advance *after* move}
7: mask $\leftarrow$ BlockingMask(o); $t \leftarrow t + 1$
8: **end while**
9: **return** metrics(t, o, info)

B. Environment

The environment operates on a 500×500 grid with Discrete(5) actions: four cardinal moves and STAY. The observation includes agent position, goal, terrain, and the current dynamic state. Within each step, the planner sees the *current* blocking mask (frozen), dynamics advance only *after* the move executes (Algorithm 1, lines 4–7). Static obstacles cause hard collisions; dynamic obstacles (fire, traffic, smoke) cause soft rejections logged via a `RejectReason` enum.

C. Dynamic Obstacles

Three dynamics layers operate simultaneously:

Fire CA. Wildfire spread follows a cellular automaton on the Moore neighborhood (8-connected). Cell states transition UNBURNED $\rightarrow$ BURNING $\rightarrow$ BURNED_OUT. When wind speed is zero, spread is isotropic with land-use-dependent probabilities (urban p=0.06, forest p=0.15, road p=0.03). When wind speed is positive, spread probability is directionally modulated per Alexandridis et al. [11]: cells downwind have increased spread probability while upwind cells are suppressed. Fire buffer (binary dilation, radius 1) blocks UAV movement within one cell of active fire. Smoke is computed via box-blur diffusion; cells with smoke ≥ 0.5 are blocked.

For corridor interdiction, fire is ignited on the A*-reference path with extended burnout (τ = 999,999 steps) ensuring corridor fires persist for the entire episode. Fig. 1 shows fire evolution on the Penteli scenario.

Traffic. N vehicles patrol road cells using greedy pathfinding. Corridor vehicles patrol near the reference corridor midpoint. Vehicle occupancy blocks UAV movement within a Manhattan buffer.

Structural Collapse. Buildings exposed to fire for τ_c=80 steps collapse, scattering permanent debris into adjacent cells (Manhattan radius 2, probability p_d=0.6). Debris is the third dynamics layer: unlike fire, which transitions to BURNED_OUT, debris is *permanent* and accumulates over the episode, progressively shrinking the navigable grid. This creates a cascading hazard coupling: fire triggers collapse, collapse spawns debris, debris blocks detour routes that planners rely on after fire corridor closure. Fig. 2 traces this cascade over a single episode: fire peaks at ∼31,000 cells (panel a), triggering 416 building collapses starting at t=85 (panel b), which scatter ∼13,000 permanent debris cells in an S-curve. Despite the growing hazard landscape, the Aggressive Replan planner navigates to the goal (panel c), illustrating that replanning frequency alone is insufficient—the planner must also read the risk map to avoid collapse zones.

D. Risk Cost Map

A continuous risk cost map $R : \mathbb{Z}^2 \rightarrow [0, 1]$ quantifies proximity to hazards. Each cell aggregates four risk layers via element-wise max:

$$R(x, y) = \max\Big(\big[1 - \tfrac{d_f}{r_f}\big]^+,\ \tfrac{s}{2},\ 0.6\big[1 - \tfrac{d_t}{r_t}\big]^+,\ 0.7\big[1 - \tfrac{d_b}{r_b}\big]^+\Big) \quad (1)$$

where d_f, d_t, d_b are Euclidean distance transforms from the fire front, traffic vehicles, and debris, r_f=30, r_t=10, r_b=8 are influence radii (cells), $s \in [0, 1]$ is smoke concentration, and $[\cdot]^+ = \max(\cdot, 0)$. Debris cells have R=1.0 (impassable) with a moderate falloff (0.7) to discourage paths near collapse zones. Each planner modulates its A*-edge cost as $w(e) = 1 + \rho \cdot R(x, y)$ using a planner-specific risk coefficient ρ:

- **A***: ignores cost map (static baseline).
- **Periodic Replan**: risk-averse, α=5.0.
- **Aggressive Replan**: risk-tolerant, β=0.5.
- **Incremental A***: moderate, γ=2.0.
- **APF**: risk-modulated repulsion, δ=3.0.

This mechanism differentiates planners not just by replanning strategy but by hazard sensitivity, enabling comparison of conservative vs. aggressive risk profiles. Fig. 3 illustrates the effect: at t=300 on Penteli, the risk-averse Periodic planner (α=5.0, green) takes a wide detour around the fire zone,

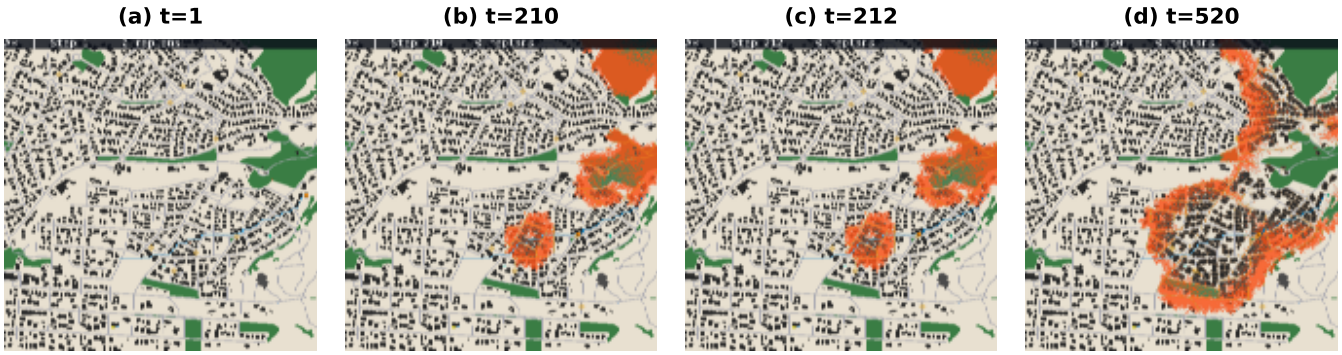

Fig. 1: Fire CA evolution on Penteli (seed 42) at event-driven timestamps. (a) t=1: corridor fires ignited on the A* reference path. (b) t=210: fire narrows corridor—last moment A* path remains passable. (c) t=212: corridor blocked—A*'s first fire rejection; adaptive planners already rerouting. (d) t=520: Incremental A* reaches goal; A* remains blocked 308 steps after corridor closure.

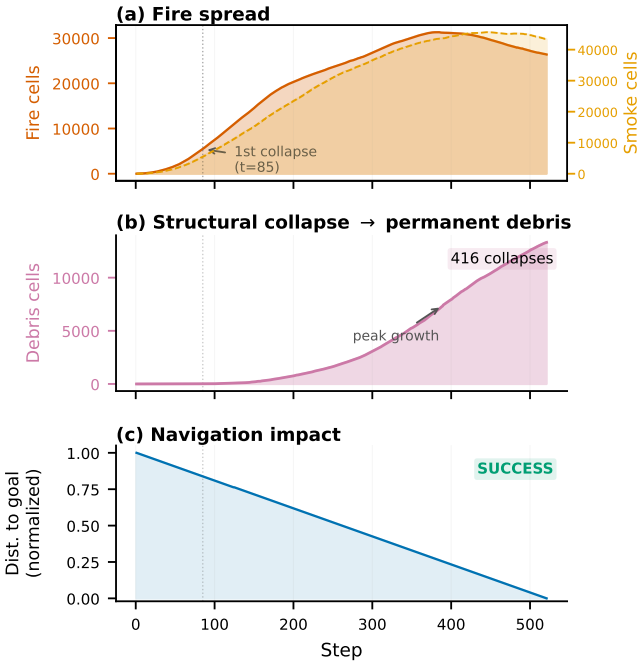

Fig. 2: Hazard cascade on Penteli (Aggressive Replan, seed 42). (a) Fire cell count and smoke diffusion. (b) Cumulative debris from 416 structural collapses; the S-curve reflects delayed collapse onset (τ_c=80 steps) followed by rapid accumulation. (c) Normalized distance-to-goal: steady decrease despite the growing obstacle field.

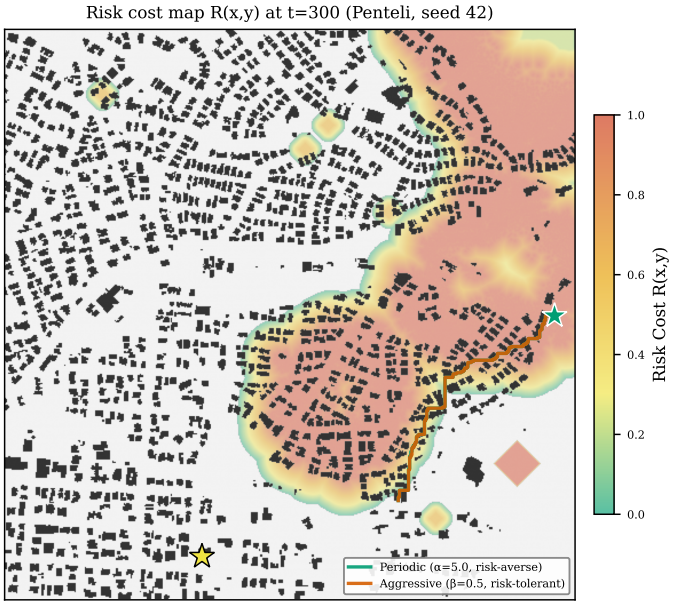

Fig. 3: Risk cost map $R(x, y)$ at t=300 (Penteli, seed 42). Color encodes aggregated risk from Eq. (1): fire proximity (red), smoke (orange), debris (pink diamonds). The risk-averse Periodic planner (green, α=5.0) detours far from fire; the risk-tolerant Aggressive planner (orange, β=0.5) follows a shorter path through the risk fringe.

while the risk-tolerant Aggressive planner (β=0.5, orange) cuts through the high-risk corridor fringe. Both reach the goal, but via qualitatively different routes.

E. Multi-POI Missions

Each scenario defines a mission with one or more *points of interest* (POIs) that the agent must visit before reaching the goal. POIs are snapped to the A*-reference corridor to preserve fairness (FC-1). The mission engine distributes multiple POIs at corridor fractions $\frac{k}{n+2}$ for k=1, . . . , n, where n is the task count.

- **Pharma delivery** (Penteli): 2 tasks (pharmacy pickup + settlement delivery), service time 1 each. *Impact:* medication efficacy $E(t)=\max(0,\ 1-(t/T)^2)$ where t is the first delivery step and $T=T_{\max}$.
- **Urban rescue** (Piraeus): 3 casualties with severity weights $w \in \{3, 2, 1\}$ (critical, serious, minor), each requiring 2 STAY steps. *Impact:* triage value $V=\sum_i w_i\, e^{-0.003\, t_i}$.
- **Fire surveillance** (Downtown): 3 survey points, each requiring 3 STAY steps. *Impact:* surveillance value $S=\sum_i(1-t_i/T)$.

Mission scores capture *when* tasks are completed, not just *whether* the agent reaches the goal. A planner that reaches the goal but skips casualties scores lower than one that rescues all three despite arriving later.

F. Scenarios

Three scenarios use pre-baked OSM tiles from Greek urban areas (Table I, Fig. 4).

TABLE I: Scenario configurations.

	Penteli	Piraeus	Downtown
Mission	Pharma	Urban rescue	Fire surv.
Tasks (service time)	2 (1)	3 (2)	3 (3)
Bldg. density	0.18	0.29	0.50
Fire ignitions	2	2	1
Vehicles	12	12	12
Max steps	2000	2000	2000

G. Planners

Five planners span three algorithmic families (Table II).

TABLE II: Planner suite with risk coefficients.

Key	Algorithm	Family	Replanning	Risk
`astar`	A*	Static	Never	—
`periodic`	A* + replan	Adaptive	Every N steps	α=5.0
`aggressive`	A* + replan	Adaptive	On mask Δ	β=0.5
`incr_astar`	Incr. A*	Adaptive	Path blocked	γ=2.0
`apf`	APF	Reactive	Every step	δ=3.0

A* [16] performs a single plan at episode start on the 4-connected grid with Manhattan heuristic. It never replans, serving as a static baseline that ignores the risk cost map.

Periodic Replan re-invokes A* every N=6 steps on the current blocking mask with risk-averse cost weighting (α=5.0).

Aggressive Replan re-invokes A* whenever the blocking mask changes near the current path, with risk-tolerant weighting (β=0.5).

Incremental A* (inspired by D* Lite [17]) uses A* with obstacle-change detection. When obstacles invalidate the current path, it replans from the current position with moderate risk weighting (γ=2.0).

APF [18] computes attractive (goal) and repulsive (obstacles within d_0=8 cells) potentials with risk-modulated repulsion (δ=3.0). When wind is active, repulsive gradients are biased toward the upwind direction, and a momentum term (α_m=0.7) smooths oscillation.

H. Contract Architecture

FLARE enforces 36 contracts across 15 families:

- **DC** (Determinism): single RNG source, bit-identical replay.
- **FC** (Fairness): corridor-aligned interdictions, identical observations.
- **EC/EV** (Events): structured reject/accept logging with authoritative step counter.
- **MP** (Mask Parity): one blocking mask function used everywhere.
- **FD/WD** (Fire/Wind): directional spread, wind determinism.
- **CC** (Calibration): feasibility pre-check, difficulty thresholds.
- **SC** (Sanity): adaptive > static in dynamic scenarios.
- **PC** (Planner): legal motion, path-progress tracking.
- **GC** (Guardrail): multi-depth relaxation for infeasible episodes.

Each contract has an associated test in the `pytest` suite (182 tests total, zero failures).

IV. Experimental Setup

A. Protocol

Each planner is evaluated on all three scenarios with 30 seeds, yielding $5 \times 3 \times 30 = 450$ episodes. Episodes where no path exists on the static grid (before dynamics) are flagged as *infeasible* and excluded from planner comparison. Infeasible episodes are uniform across planners (7 per planner, 7.8% exclusion rate), confirming fairness contract FC-2.

B. Ablation Studies

Three ablation studies isolate individual design factors: *(i)* dynamics isolation (fire/traffic/all, 450 episodes), *(ii)* replan frequency sweep (cadences 3–48, 150 episodes), *(iii)* fire intensity sweep (1–8 ignition points, 250 episodes).

C. Metrics

- **Success rate** (SR): fraction of feasible episodes where the agent reaches the goal within the horizon.
- **Mission score** (MS): time-weighted task-completion score (§III-E). Higher is better.
- **Tasks completed**: number of POIs serviced before episode end (out of 2 or 3 per scenario).
- **Path length**: cells traversed (successful episodes only).
- **Replans**: number of path recomputations.

D. Statistical Methods

Following Agarwal et al. [10], we report IQM with 95% bootstrap CIs (10,000 resamples). Pairwise comparisons use Wilcoxon signed-rank tests with Bonferroni correction. Cliff's delta quantifies effect sizes. Performance profiles show the fraction of episodes where each planner is within $\tau\times$ the best path length [13].

V. Results

A. Main Results

Table III summarizes results across all 450 episodes. Of 90 episodes per planner, 7 (7.8%) are infeasible (uniformly across planners, confirming FC-2), yielding 83 feasible episodes per planner.

A* achieves 0% success: fire corridor closures and debris block the reference path, and the static planner cannot deviate (Fig. 5). Incremental A* leads at 74.7% on feasible episodes, followed by Periodic Replan (73.5%) and Aggressive Replan (61.4%). APF drops to 47.0%, severely impacted by debris-induced local minima.

Fig. 4: Three OSM-based scenarios (basemap at t=0, no dynamics). Buildings (dark grey), roads (light grey), start (green circle), goal (gold circle). Building density increases left to right (0.18, 0.29, 0.50), reducing available detour routes.

TABLE III: Overall planner comparison (450 episodes, 30 seeds $\times$ 3 scenarios). SR_{all} includes infeasible episodes; SR_{feas} excludes them. Path length: successful only; Steps/Replans/Time: all feasible episodes.

Planner	SR_{all} (%)	SR_{feas} (%)	Path (cells)	Steps	Replans	Time (s)
A*	0.0	0.0	—	646	0	40
Periodic	67.8	73.5	383	650	70.6	44
Aggressive	56.7	61.4	386	643	119.8	41
Incr. A*	68.9	74.7	356	541	3.3	34
APF	43.3	47.0	487	806	16.1	50

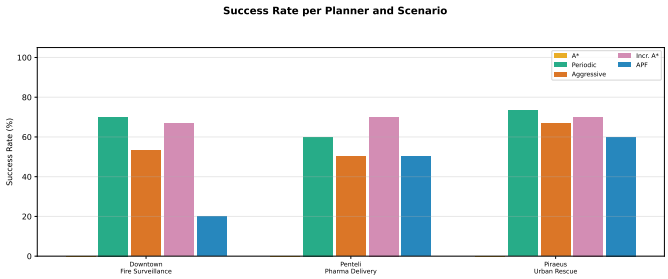

Fig. 5: Success rate per planner and scenario. A* fails uniformly. APF is most affected by high-density Downtown (23%), where debris creates local-minima traps.

Incremental A* navigation dominance. Incremental A* achieves the highest feasible SR (74.7%) with only 3.3 replans—the most efficient replanner—and fastest computation (34s). Its graph-repair strategy avoids unnecessary replanning while maintaining obstacle awareness. Periodic Replan follows at 73.5% with moderate replans (70.6) and reasonable computation time (44s). However, Incremental A*'s minimal replanning causes it to skip mission tasks (Sec. V-B). Fig. 6 compares all five planners on the same episode.

APF limitations. APF achieves the lowest success rate (47.0% feasible) and produces paths 30% longer than graph-based planners (487 vs. 375 cells). Its reactive gradient-descent approach creates suboptimal paths and is severely susceptible to local minima when debris creates permanent dead ends.

B. Mission Impact

Table IV shows mean mission scores per planner and scenario. Unlike success rate—a binary outcome—mission scores capture the *quality* of task completion.

TABLE IV: Mean mission score by planner and scenario (30 seeds). Penteli: medication efficacy [0–1]; Piraeus: triage value [0–6]; Downtown: surveillance freshness [0–3]. Higher is better. **Bold**: best per scenario.

Planner	Penteli (efficacy)	Piraeus (triage)	Downtown (surv.)
A*	0.00	0.00	0.00
Periodic	0.93	2.89	1.54
Aggressive	**0.99**	**2.98**	**1.68**
Incr. A*	0.40	1.37	0.72
APF	0.43	1.55	0.87

Mission scores reveal a *ranking inversion*: Incremental A* achieves the *highest* SR (74.7%) but ranks *last* among adaptive planners in mission score. On Downtown, Incremental A* completes only 0.8 of 3 surveillance tasks (score 0.72), while Aggressive Replan completes 1.9 tasks (score 1.68) despite lower SR (61.4%). The mechanism is clear: Incremental A* replans only when blocked (3.3 replans), rushing to the goal and bypassing intermediate POIs. Aggressive Replan's frequent replanning (120 replans) exposes it to more tasks along alternative routes.

On Piraeus (urban rescue, 3 victims with triage weights 3/2/1), Aggressive Replan again leads (2.98 vs. 1.37 for Incremental A*), rescuing 2.2 victims vs. 1.0. A planner that optimizes for goal-reaching alone misses the human cost of bypassed victims.

Fig. 7 illustrates this inversion on a single Piraeus episode (seed 11, 3 victims scattered off-corridor). A* gets stuck (SR 0%, score 0.00, 0/3 tasks). Incremental A* reaches the

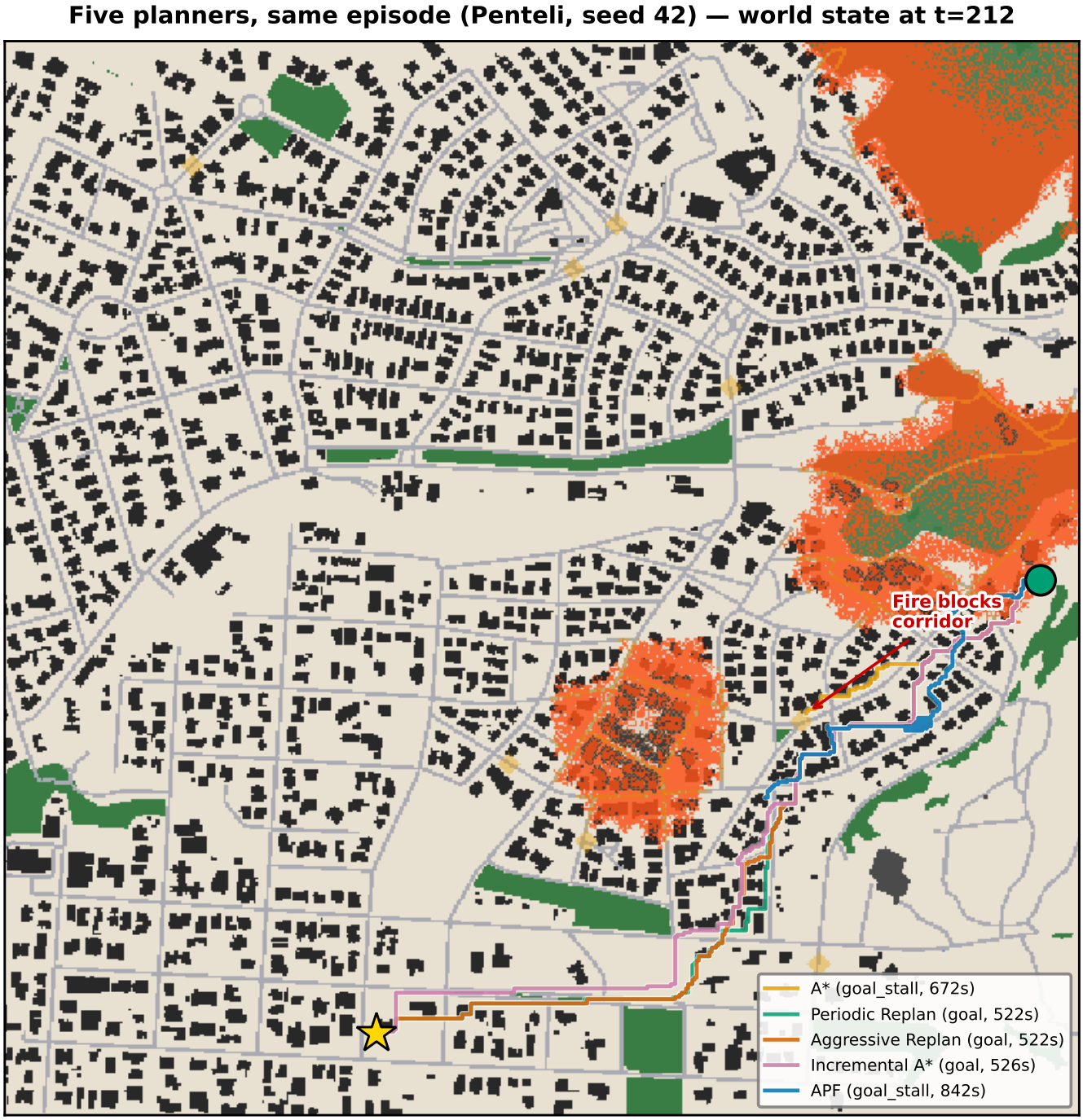

Fig. 6: All five trajectories on a single map (Penteli, seed 42). A* (orange) is blocked at the corridor and oscillates in place. Graph-based adaptive planners (green, red, pink) detour around the fire zone and reach the goal at $t \approx 522$. APF (blue) follows a longer, erratic path and times out, trapped by fire-front local minima. Annotation marks the divergence point where fire blocks the corridor.

goal in 141 steps but bypasses all three victims (SR 100%, score 0.00). Aggressive Replan visits 2 of 3 victims *and* reaches the goal in 353 steps (SR 100%, score 3.64)—the highest mission score of any planner.

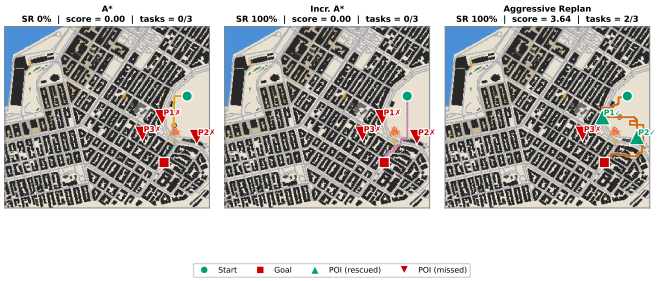

Fig. 7: Ranking inversion on Piraeus (seed 11). Three panels show A*, Incr. A*, and Aggressive Replan on the same episode with 3 victims (P1–P3). Incr. A* achieves 100% SR but scores 0.00 (no victims rescued); A* scores 0.00 (stuck); Aggressive Replan scores 3.64 (2/3 tasks rescued).

Fig. 8 plots SR vs. normalized mission score per planner across all scenarios, showing the trade-off between navigation success and mission impact at a glance.

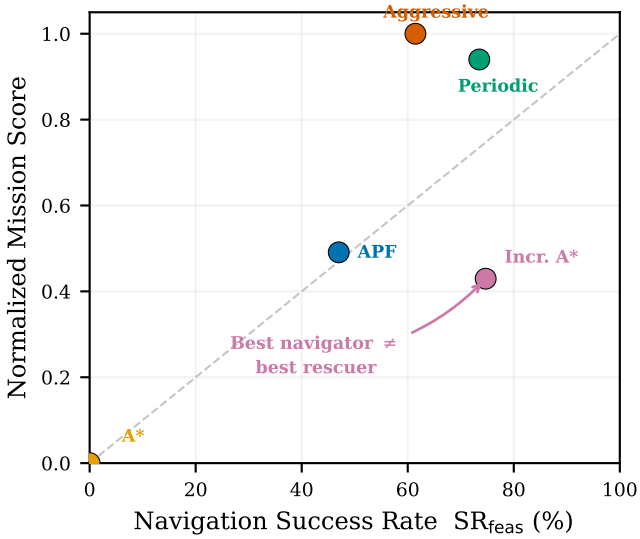

Fig. 8: Success rate vs. normalized mission score. Ideal planners occupy the upper-right corner. Aggressive Replan achieves the best mission score despite lower SR than Periodic; Incr. A* underperforms on mission quality despite strong navigation.

C. Statistical Analysis

Effect sizes. Cliff's delta between A* and each adaptive planner ranges from $|d| = 0.47$ (APF, medium) to $|d| = 0.75$ (Incr. A*, large). Between adaptive planners, Periodic vs. APF yields $d = +0.27$ (small); Incr. A* vs. APF yields $d = +0.28$ (small); other pairwise deltas are negligible ($|d| < 0.15$).

Significance. The Friedman test on success rate across 83 feasible (scenario, seed) pairs is significant: $\chi^2(4) = 127.5$, $p < 0.001$. Wilcoxon pairwise post-hoc tests (Bonferroni corrected, k=10) show: A* vs. all adaptive planners $p < 0.001$; APF vs. Periodic $p = 0.009$; APF vs. Incr. A* $p = 0.006$; no significant difference between Periodic and Aggressive ($p = 0.25$), Periodic and Incr. A* ($p = 1.0$), or Aggressive and APF ($p = 0.58$).

Mission score Friedman. A separate Friedman test on mission scores is also significant: $\chi^2(4) = 207.5$, $p < 0.001$, confirming that planners differ not only in whether they reach the goal but in *how much human value* they deliver en route.

D. Replanning Behavior

Fig. 9 compares episode timelines. Aggressive Replan triggers the most replans (mean 120); Periodic triggers fewer (mean 71) at fixed intervals; Incremental A* is the most conservative (mean 3.3), replanning only when blocked—36× fewer replans yet the highest SR (74.7%).

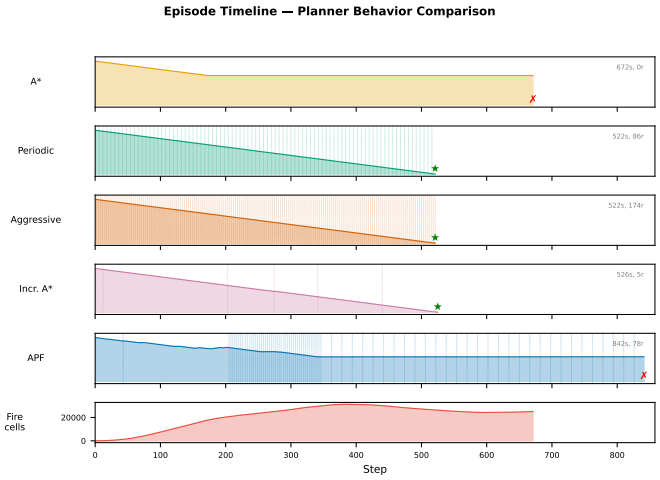

Fig. 9: Episode timeline for Penteli (seed 42). Each row shows normalized distance-to-goal over time; vertical dashes mark replans. Bottom panel: fire cell count growth. Periodic, Aggressive, and Incremental A* reach the goal in ∼522 steps; A* times out (blocked by corridor fire at step 212); APF also times out after oscillating near fire fronts.

E. Ablation: Dynamics Isolation

Table V isolates each dynamics layer (10 seeds × 3 scenarios = 30 episodes per cell). Fire is the primary differentiator: A* drops to 0% while graph-based adaptive planners achieve 73–83%. Traffic alone reduces A* to 7% but adaptive planners maintain 97%. APF degrades most under combined dynamics (40%) and fire-only (43%), reflecting its susceptibility to local-minima traps near expanding fire fronts.

The replan-frequency sweep (cadences 3–48) shows SR peaks at 77% for cadence 3, declining to 57% at cadence 48. The fire-intensity sweep (1–8 ignitions) confirms that adaptive planners maintain 60–100% SR across all intensities while A* remains at 0%; Incremental A* peaks at 100% with 4 ignitions and the overall adaptive advantage persists even at 8 simultaneous fire sources.

TABLE V: Ablation: success rate (%) by dynamics layer (10 seeds, 3 scenarios).

Planner	Fire	Traffic	All
A*	0	7	0
Periodic	73	97	73
Aggressive	83	97	77
Incr. A*	80	97	80
APF	43	97	40

VI. Discussion

The replanning gap. Planners that replan succeed; the static baseline fails entirely. Fire is the primary cause (A* 0% vs. adaptive 73–83% in fire-only ablation).

Incremental A* navigation efficiency. Incremental A* achieves the highest feasible SR (74.7%) with only 3.3 replans, the most efficient replanner. Periodic Replan follows at 73.5% with moderate overhead (71 replans). Yet Incremental A* bypasses mission tasks (Sec. V-B).

Mission scores expose hidden trade-offs. Multi-POI scores (Table IV, Fig. 7) reveal ranking inversions invisible to SR alone. Incremental A* (74.7% SR) scores *last* among adaptive planners because minimal replanning bypasses off-corridor POIs. Aggressive Replan yields the highest triage value (2.98) despite lower SR than Periodic (61% vs. 74%), because frequent replanning (β=0.5) exposes it to more POIs (Fig. 8). The *best navigation planner* is not necessarily the *best mission planner*.

Limitations. FLARE operates on 2D grids without altitude or terrain slope [22]. Risk coefficients are not optimized per-scenario. Future directions include altitude, multi-agent coordination, and Bayesian risk-coefficient tuning.

VII. Conclusion

FLARE provides a reproducible, contract-verified benchmark for UAV path planning in disaster-response scenarios. Our evaluation of five planners across 450 episodes demonstrates that (1) replanning capability is necessary for navigating fire-and-debris-affected terrain (A* 0% vs. adaptive 47–75%), (2) incremental graph repair (Incr. A*, 75%) achieves the highest navigation SR but bypasses mission tasks, while (3) mission-impact scores reveal ranking inversions where the best navigation planner is not the best disaster-response planner. The framework's 36 contracts, multi-POI missions, and time-decaying scores provide a substrate for evaluating planners with human consequences in mind. Code: https://github.com/zervakisai/uavplan (MIT license).

Acknowledgments

Map data © OpenStreetMap contributors, available under the Open Database License (ODbL). Scenarios are inspired by the 2021 Evia wildfires, 2023 Rhodes coastal fires, and 2023 Evros megafire.

References

[1] M. Erdelj, E. Natalizio, K. R. Chowdhury, and I. F. Akyildiz, "Help from the sky: Leveraging UAVs for disaster management," *IEEE Pervasive Computing*, vol. 16, no. 1, pp. 24–32, 2017.

[2] C. Yuan, Y. Zhang, and Z. Liu, "A survey on technologies for automatic forest fire monitoring, detection, and fighting using unmanned aerial vehicles and remote sensing techniques," *Canadian Journal of Forest Research*, vol. 45, no. 7, pp. 783–792, 2015.

[3] G. Yücesoy *et al.*, "The role of drones in disaster response: A literature review of operations research applications," *International Transactions in Operational Research*, 2025.

[4] Z. Siddiqui *et al.*, "A systematic literature review of drones in emergency medicine: Practical applications, legal challenges, and future directions," *Drone Systems and Applications*, 2024.

[5] A. Moustafa *et al.*, "Advances in UAV path planning: A comprehensive review of methods, challenges, and future directions," *Drones*, vol. 9, no. 5, p. 376, 2025.

[6] A. Dradoum, A. Khelassi, and F. Lachekhab, "Intelligent path planning algorithms for UAVs: Classification, complexity analysis, hybrid ablation insights, and future directions," *Advances in Mechanical Engineering*, 2025.

[7] S. Aggarwal and N. Kumar, "Path planning techniques for unmanned aerial vehicles: A review, solutions, and challenges," *Computer Communications*, vol. 149, pp. 270–299, 2020.

[8] Y. Zhao, Z. Zheng, and Y. Liu, "Survey on computational-intelligence-based UAV path planning," *Knowledge-Based Systems*, vol. 158, pp. 54–64, 2018.

[9] P. Henderson, R. Islam, P. Bachman, J. Pineau, D. Precup, and D. Meger, "Deep reinforcement learning that matters," in *Proceedings of the 32nd AAAI Conference on Artificial Intelligence*, 2018.

[10] R. Agarwal, M. Schwarzer, P. S. Castro, A. Courville, and M. G. Bellemare, "Deep reinforcement learning at the edge of the statistical precipice," in *Advances in Neural Information Processing Systems*, vol. 34, 2021.

[11] A. Alexandridis, D. Vakalis, C. I. Siettos, and G. V. Bafas, "A cellular automata model for forest fire spread prediction: The case of the wildfire that swept through Spetses island in 1990," *Applied Mathematics and Computation*, vol. 204, no. 1, pp. 191–201, 2008.

[12] J. Demšar, "Statistical comparisons of classifiers over multiple data sets," *Journal of Machine Learning Research*, vol. 7, pp. 1–30, 2006.

[13] E. D. Dolan and J. J. Moré, "Benchmarking optimization software with performance profiles," *Mathematical Programming*, vol. 91, no. 2, pp. 201–213, 2002.

[14] M. Towers, A. Kwiatkowski, J. K. Terry, J. U. Balis, G. De Cola, T. Deleu, M. Goulão *et al.*, "Gymnasium: A standard interface for reinforcement learning environments," in *Advances in Neural Information Processing Systems*, vol. 38, 2025, spotlight.

[15] G. Boeing, "OSMnx: New methods for acquiring, constructing, analyzing, and visualizing complex street networks," *Computers, Environment and Urban Systems*, vol. 65, pp. 126–139, 2017.

[16] P. E. Hart, N. J. Nilsson, and B. Raphael, "A formal basis for the heuristic determination of minimum cost paths," *IEEE Transactions on Systems Science and Cybernetics*, vol. 4, no. 2, pp. 100–107, 1968.

[17] S. Koenig and M. Likhachev, "D* Lite," in *Proceedings of the 18th AAAI Conference on Artificial Intelligence*, 2002, pp. 476–483.

[18] O. Khatib, "Real-time obstacle avoidance for manipulators and mobile robots," *The International Journal of Robotics Research*, vol. 5, no. 1, pp. 90–98, 1986.

[19] W. Zhou *et al.*, "A risk-based unmanned aerial vehicle path planning scheme for complex air–ground environments," *Risk Analysis*, 2025.

[20] C. Melissaris *et al.*, "Risk-aware UAV trajectory optimization using open urban GIS data and target level of safety constraints," *Drones*, vol. 9, no. 10, p. 666, 2025.

[21] F. Bonsignorio *et al.*, "Towards a science of reproducible results in robotics," *Nature Machine Intelligence*, 2025.

[22] F. Giannino *et al.*, "Statistical dynamics of wildfire burned area from cellular-automata simulators," *Scientific Reports*, vol. 15, 2025.

[23] M. Ferrara *et al.*, "Patterns of burned area by a cellular-automata fire simulator: The role of microscale wind field," *Science China Earth Sciences*, 2025.